
Business Requirements Document (BRD)

1. Document Overview

- **Project Name:** Murex Real-Time & End-of-Day XVA Integration Platform
- **Document Type:** Business Requirements Document (BRD)
- **Scope:** Real-Time Pre-Trade Pricing (Artillium), End-of-Day Aggregations (CVA Viewer), Upstream Feeds Ingestion, and Murex Configuration Modules.

2. Business Problem & Solution Summary

2.1 Core Business Problem

Front Office trading desks book derivative transactions without factoring in real-time counterparty credit risks and funding costs. This causes financial exposure and operational failures:

- **Unadjusted Fair Value Pricing:** Base Net Present Value (NPV) does not capture credit risk or funding costs, leading to unpriced counterparty defaults.
- **Disconnected Upstream Static Data:** Missing or incorrect parent-child counterparty hierarchies, unlinked legal agreements (netting/CSA), or outdated credit ratings distort Loss Given Default (LGD) and risk valuations.
- **Engine & Infrastructure Failures:**
 - Time zone differences cause desk dates to roll forward past midnight before live pricing engines reset, resulting in blank pricing screens.
 - Heavy compute loads during concurrent Sunday engine runs trigger Garbage Collection (GC) memory spikes and crashes.
 - Cloud dynamic scale-downs (AWS) break dynamic features like interactive "What-If" portfolio analysis.

2.2 Proposed Solution

Implement a end-to-end Murex XVA solution covering live pre-trade pricing, batch reporting, and automated static data pipelines:

- **Mandatory Pre-Trade Pricing:** Force pre-trade XVA pricing on the Trade Pad via the **Artillium Real-Time Engine** to apply CVA/FVA premiums before deal acceptance.
- **Automated Data Pipelines:** Synchronize daily feeds from SAP (ratings), GoldenSource/Markit (credit curves), and GMDI (netting/CSA) into Murex/MLC static tables.
- **System Stability & Configuration Control:** Implement Free Trade Rules for date mismatches, optimize JVM memory during concurrent engine runs, and control AWS auto-scaling.

3. Murex Screens & Technical Interfaces Used

The table below outlines the primary Murex screens, menus, and interfaces utilized across the system workflow:

Murex Screen / Module Name	Purpose / Functionality	Key Inputs & Fields Displayed
E-Trade Pad (E-Pad)	Used by Front Office traders to input, price, and book trades. Enforces mandatory XVA pricing prior to booking.	Base NPV, Adjusted NPV, Broker Flows (CVA/FVA Premiums).
XVA Preview Screen	Accessible via trade pad buttons (Pricing / Preview) to view detailed trade exposure profiles and portfolio impacts.	Effective Exposure Profile Graph, Simulated MV vs. Front Office MV, Incremental XVA, Parent Counterparty, Netting & CSA Agreement flags.
Credit Curve Screen	View market quotes and computed credit probabilities for specific counterparties.	Market Quotes (e.g., 6M, 10Y), Non-Default Probability, Default Probability (\$1 - \text{Non-Default}\$).
CVA Model Settings Menu	Configuration menu used to define pricing models and market data scopes for real-time and batch engines.	Model Selection (e.g., Artillium), Scope Currencies, Rate/Equity/Commodity Instruments, Market Data Closing Sources (e.g., Sydney Closing).
Formulae Config Menu	Configuration area defining mathematical logic for risk calculation and aggregation.	Formula definitions (e.g., Netting Group XVA) used by MLC/XVA aggregation rules.
MLC Static Data Screen	Replicated static database holding counterparty risk attributes synchronized from Murex Core DB.	XVA Eligible User-Defined Flag (UDF), Counterparty Parent-Child Hierarchies, Rating Mappings.
Scope Configuration / Data Mart	Filters trades eligible for real-time valuation or end-of-day reports.	Portfolio Eligibility UDFs (Yes/No), Trade Typology scoping, PL-Accepted status.
CVA Viewer	Main risk management screen used by PCVA desks to view aggregated portfolio risk reports.	Summary Reports (Parent Counterparty level), Attribution Reports (Trade level), Sensitivity Bumps (+1 bps), "What-If" Analysis.
Lookup Tables Screen	Administrative control tables for operational overrides and typology scoping.	CVA Typologies (In/Out of scope), Skip Mandatory Pricing table (Yes/No override by user group).
GMP Tables	System storage for risk correlation parameters used in advanced risk analysis.	Correlation matrices between market risk factors (e.g., crude oil prices) and counterparty credit risks.

4. Detailed Business Requirements

4.1 Functional Requirements

- **Mandatory Pre-Trade Pricing (Artillium Real-Time Engine):**
 - Traders must trigger mandatory pricing on the **E-Trade Pad** prior to deal execution.
 - System must capture CVA and FVA premiums, recording them as broker flows to adjust the raw NPV.
 - Allow authorized groups (MarketWire, AllInfo, CVA Desk) to bypass mandatory pricing using the **Skip Mandatory Pricing Lookup Table** during engine outages.
 - Force dynamic pre-trade re-pricing on event processing screens during trade restructurings or amendments.
- **Upstream Data Ingestion & Data Pipelines:**
 - **Credit Curves:** Ingest vendor quotes daily from GoldenSource/Markit into the **Credit Curve Screen** to calculate default and non-default probabilities.
 - **Credit Ratings:** Import counterparty credit ratings daily from SAP during EOD processing to assign Loss Given Default (LGD) and recovery rates.
 - **Legal Agreements:** Ingest netting and collateral (CSA) legal agreements daily from GMDI to discount potential exposures.
 - **MLC Data Sync:** Run automated daily EOD scripts to sync counterparty User-Defined Fields (UDFs) such as XVA Eligible into the **MLC Static Data Screen**.
- **End-of-Day Aggregation & Risk Reporting (CVA Viewer Engine):**
 - Generate trade-level **Attribution Reports** and parent-level **Summary Reports** within the **CVA Viewer** for PL-accepted trades.
 - Compute +1 bps interest rate bump sensitivity reports to forecast market movement impacts.
 - Run weekly Wrong-Way Risk (WWR) correlation checks stored in **GMP Tables** to evaluate correlations between market factors and default risks.
 - Maintain interactive "What-If" portfolio analysis capabilities on the **CVA Viewer**.

4.2 Non-Functional & System Requirements

- **Engine Reset & Date Alignment:** Implement Free Trade Rules or alert warnings when user desk dates roll forward past midnight before the scheduled 3:00 AM **Artillium Engine (RTM)** reset.
- **Memory & Infrastructure Optimization:** Optimize JVM Garbage Collection (GC) parameters to prevent server crashes during concurrent Sunday runs of the **Artillium** and **CVA Viewer** engines.
- **Cloud Resource Scaling:** Maintain static compute node allocations during business hours and EOD runs in AWS to prevent functional breaks in **CVA Viewer "What-If"** sessions.

5. Traceability Matrix

Ref #	Identified Business Problem	Proposed Solution	Murex Module / Screen Involved
GAP-01	Unpriced credit/funding risk during trade execution.	Mandate pre-trade XVA pricing to calculate CVA/FVA premiums before trade booking.	E-Trade Pad, XVA Preview Screen, Artillium Engine
GAP-02	Inaccurate valuations from missing ratings or legal structures.	Automate daily EOD ingestion scripts from SAP, GMDI, and GoldenSource into MLC tables.	MLC Static Data, Credit Curve Screen, SAP/GMDI Interfaces
GAP-03	Blank pricing screens post-midnight due to date mismatches.	Add Free Trade Rules and adjust engine reset cycles relative to desk dates.	E-Trade Pad, Artillium RTM Engine
GAP-04	"What-If" CVA Viewer features breaking during cloud scale-downs.	Stabilize AWS scaling settings to maintain active nodes during batch/user sessions.	CVA Viewer Screen, AWS Infrastructure
GAP-05	GC memory spikes and engine crashes during concurrent Sunday runs.	Tune JVM GC parameters and decouple execution schedules for batch and real-time engines.	Core Server LTS Scripts, Artillium & CVA Viewer Engines